

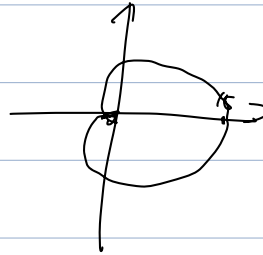
Nyquist

$$L(s) = \frac{90}{(s+3)(s+6)} = \frac{15}{(1+\frac{s}{3})(1+\frac{s}{6})}$$

$$\arg[L(s)] = -\tan^{-1}\left(\frac{\omega}{3}\right) - \tan^{-1}\left(\frac{\omega}{6}\right)$$

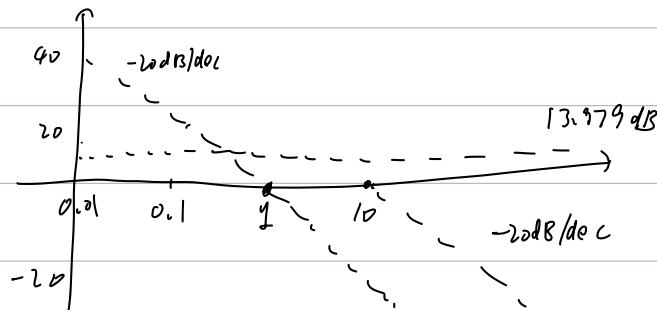
$$\omega=0 \Rightarrow \arg = 0 \quad 15$$

$$\omega=\infty \Rightarrow \arg = -180^\circ \quad 0$$



Q3. (a). $G(s) = \frac{5}{j\omega(1+\frac{\omega}{10}j)}$

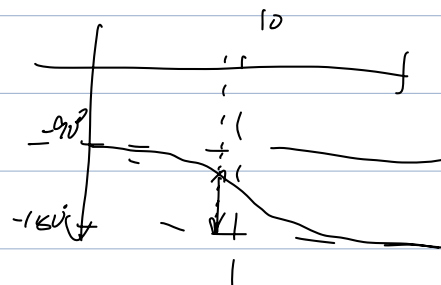
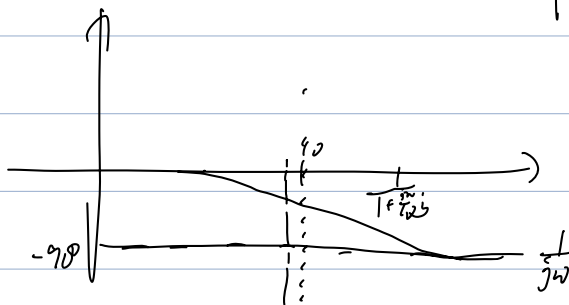
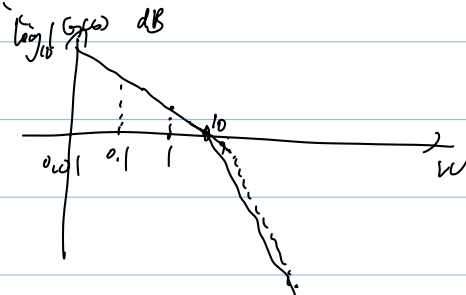
(b) $20\log|G(s)| = 20\log_0 5 - 20\log_0 \omega - 10\log_0(1+\frac{\omega^2}{100})$



$0 \sim 10$ -20 dB/dec

10 $-3 \text{ dB compensation}$

$10 \sim \infty$ -40 dB/dec



Routh Hurwitz
Nyquist stability criterion
Root locus analysis